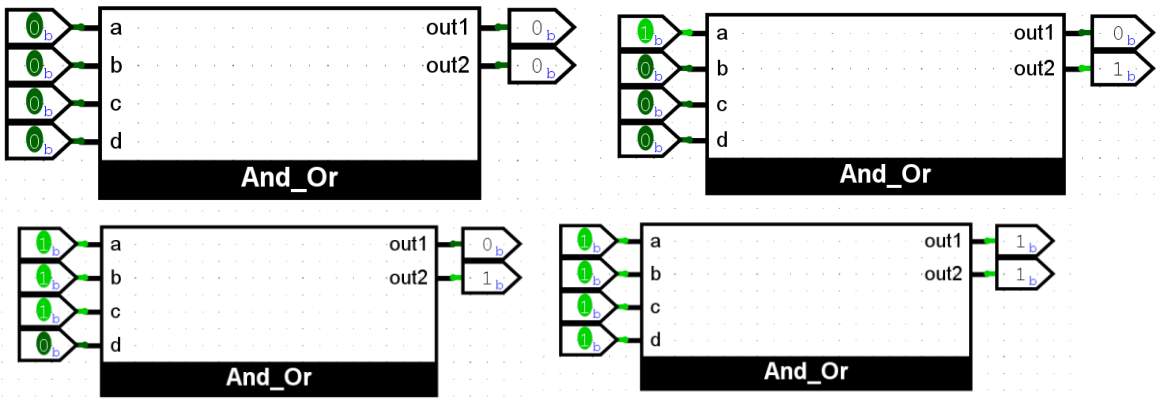
(VHDL) Assignment-2 2023050273 Jo Suhyeok

- Problem 1 (And_Or)
  - VHDL Code

```
1  --library and package
2  LIBRARY ieee;
3  USE ieee.std_logic_1164.all;
4  USE ieee.std_logic_arith.all;
5  USE ieee.std_logic_unsigned.all;
6
7
8  --entity (input&output)
9  ENTITY And_Or IS
10      PORT (
11          a,b,c,d         :IN  std_logic;
12          out1,out2       :OUT std_logic
13      );
14 END And_Or;
15
16 --architecture
17 ARCHITECTURE bhv OF And_Or IS
18 BEGIN
19
20      out1 <= a and b and c and d;
21      out2 <= a or b or c or d;
22 END ARCHITECTURE bhv;
```

  - Circuit simulation screenshots

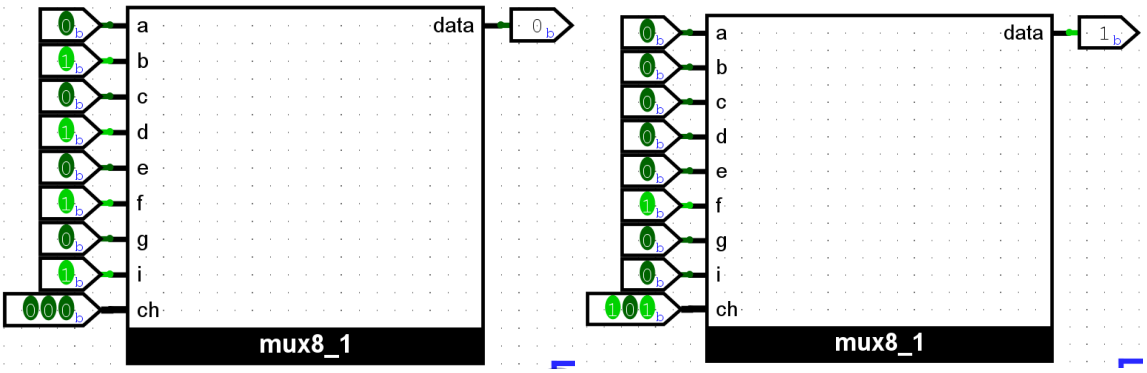


  - Out1 <= a and b and c and d
    Out2 <= a or b or c or d
    If a is 1 and else is 0, out1 becomes 1. If all of the inputs are 1, out2 becomes 1. Therefore 'out1' output means the AND gate, and 'out2' output means the OR gate.

- Problem 2 (mux8_1)
  - VHDL Code

```
2  LIBRARY ieee;
3  USE ieee.std_logic_1164.all;
4  USE ieee.std_logic_arith.all;
5  USE ieee.std_logic_unsigned.all;
6
7
8  --entity (input&output)
9  ENTITY mux8_1 IS
10      PORT (
11          a,b,c,d,e,f,g,i      :IN  std_logic;
12          ch                   :IN  std_logic_vector(2 downto 0);
13          data                 :OUT std_logic
14      );
15 END mux8_1;
16
17 --architecture
18 ARCHITECTURE bhv OF mux8_1 IS
19 BEGIN
20      process(a,b,c,d,e,f,g,i,ch)
21      begin
22          case ch is
23              when "000" => data <= a;
24              when "001" => data <= b;
25              when "010" => data <= c;
26              when "011" => data <= d;
27              when "100" => data <= e;
28              when "101" => data <= f;
29              when "110" => data <= g;
30              when "111" => data <= i;
31              when others => data <= '0';
32          end case;
33      end process;
34 END ARCHITECTURE bhv;
```

  - Circuit simulation screenshots



  - The 'ch' input represents the channel and determines which input among a, b, c, d, e, f, g, and i is output as data through the channel. Therefore, when ch is 000, input a is output, and when ch is 101, input f is output.

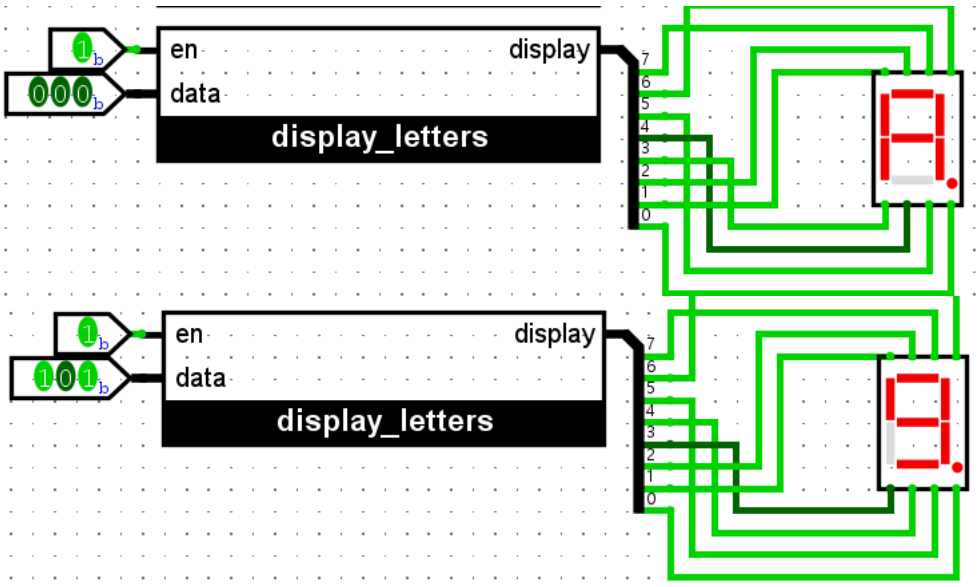
- Problem 3 (display letters)
  - VHDL Code

```vhdl
--library and package
LIBRARY ieee;
USE ieee.std_logic_1164.all;
USE ieee.std_logic_arith.all;
USE ieee.std_logic_unsigned.all;

--entity (input&output)
ENTITY display_letters IS
     PORT (
          en             :IN  std_logic;
          data           :IN  std_logic_vector(2 downto 0);
          display        :OUT std_logic_vector(7 downto 0)
     );
END display_letters;

--architecture
ARCHITECTURE bhv OF display_letters IS
BEGIN
     process(en,data)
     begin
          if en='0' then
               display <= "ZZZZZZZZ";
          else
               case data is
                    when "000" => display <= "11101111";
                    when "001" => display <= "00111111";
                    when "010" => display <= "00011011";
                    when "011" => display <= "01111011";
                    when "100" => display <= "10001111";
                    when "101" => display <= "11110111";
                    when others => display <= "00000000";
               end case;
          end if;
     end process;
END ARCHITECTURE bhv;
```

  - Circuit simulation screenshots



  - This VHDL code displays alphabetic characters on a 7-segment display according to the data value. When en is 0, all segments of the 7-segment display are turned off.

- Problem 4 (vote25)

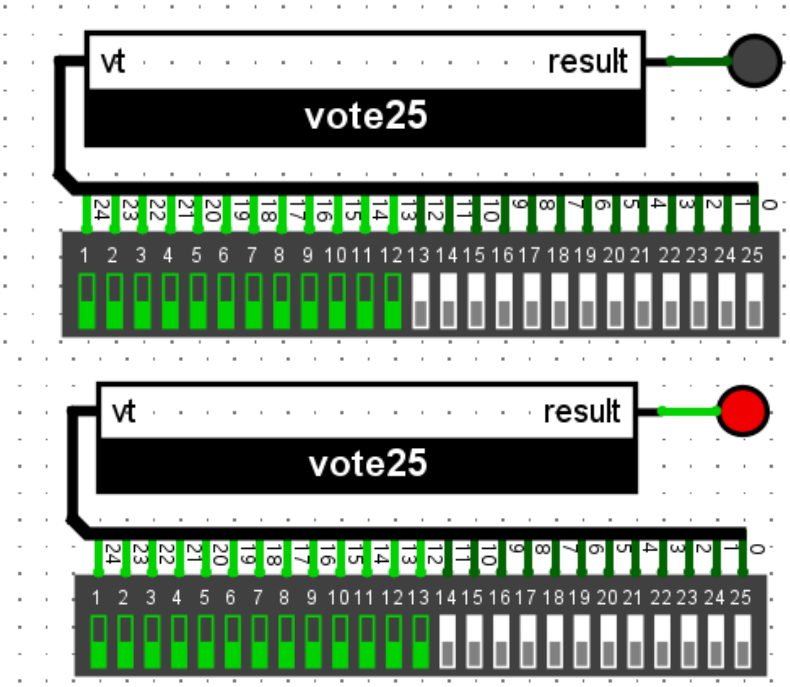
■ VHDL Code

```vhdl
--library and package
LIBRARY ieee;
USE ieee.std_logic_1164.all;
USE ieee.std_logic_arith.all;
USE ieee.std_logic_unsigned.all;

--entity (input&output)
ENTITY vote25 IS
     PORT (
          vt             :IN  std_logic_vector(25 downto 1);
          result         :OUT std_logic
     );
END vote25;

--architecture
ARCHITECTURE bhv OF vote25 IS
BEGIN
     process(vt)
     variable sum:integer range 0 to 25;
     begin
          sum:=0;
          for i in 1 to 25 loop
               if (vt(i)='1') then
                    sum:=sum+1;
               end if;
          end loop;
          if sum >= 13 then
               result <= '1';
          else
               result <= '0';
          end if;
     end process;
END ARCHITECTURE bhv;
```

■ Circuit simulation screenshots



■ This VHDL code turns on the switch when the number of bits set to '1' in the 25-bit input vt is 13 or more.

- Problem 5 (probelm5

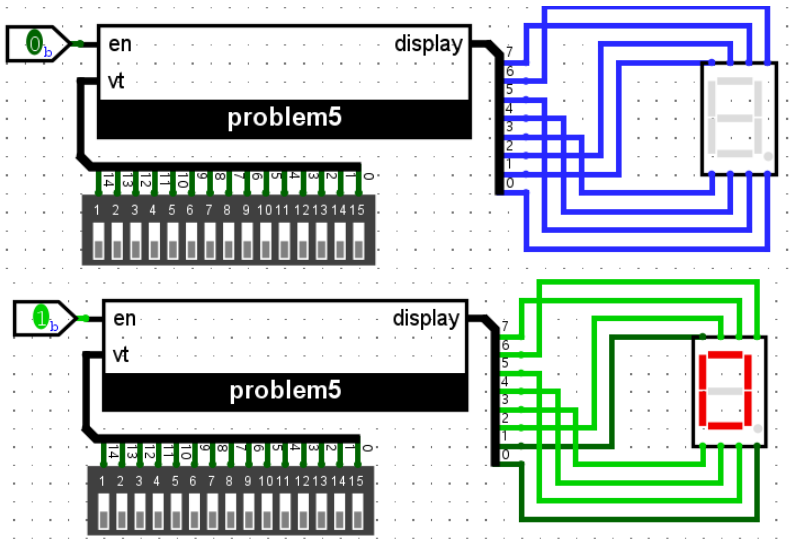
■ VHDL Code

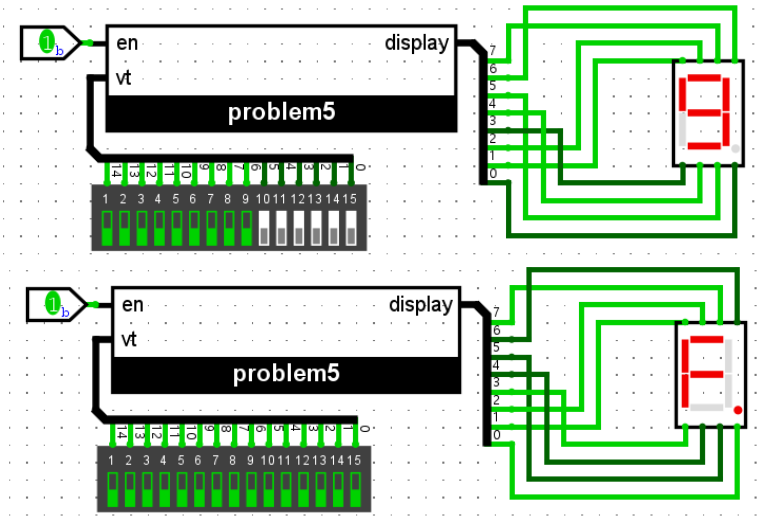
```
--library and package
LIBRARY ieee;
USE ieee.std_logic_1164.all;
USE ieee.std_logic_arith.all;
USE ieee.std_logic_unsigned.all;

--entity (input&output)
ENTITY problem5 IS
     PORT (
          en             :IN  std_logic;
          vt             :IN  std_logic_vector(14 downto 0);
          display        :OUT std_logic_vector(7 downto 0)
     );
END problem5;
```

```
ARCHITECTURE bhv OF problem5 IS
BEGIN
     process(en,vt)
     variable sum:integer range 0 to 15;
     begin
          if en='0' then
               display <= "ZZZZZZZZ";
          else
               sum:=0;
               for i in 0 to 14 loop
                    if (vt(i)='1') then
                         sum:=sum+1;
                    end if;
               end loop;
               case sum is
                    when 0 => display <= "11111100";
                    when 1 => display <= "01100000";
                    when 2 => display <= "11011010";
                    when 3 => display <= "11110010";
                    when 4 => display <= "01100110";
                    when 5 => display <= "10110110";
                    when 6 => display <= "10111110";
                    when 7 => display <= "11100100";
                    when 8 => display <= "11111110";
                    when 9 => display <= "11110110";
                    when 10 => display <= "11101111";
                    when 11 => display <= "00111111";
                    when 12 => display <= "00011011";
                    when 13 => display <= "01111011";
                    when 14 => display <= "10011111";
                    when 15 => display <= "10001111";
                    when others => display <= "00000000";
               end case;
          end if;
     end process;
END ARCHITECTURE bhv;
```

■ Circuit simulation screenshots





- This VHDL code counts the number of bits set to '1' in the 15-bit input vt. If the count is less than 10, it displays the corresponding number. If the count is between 10 and 15, it displays the corresponding alphabetic character from A to F, respectively.